DATABASE SYSTEMS COMPREHENSIVE - ESSAY SECTION

Q1. Explain the ACID properties of relational database management systems and contrast them with eventual consistency models. [10 Marks]

Marking Rubric:

- Atomicity definition and transaction rollback explanation (2 marks)
- Consistency preservation and constraint invariants (2 marks)
- Isolation levels (Read Committed, Serializable) and concurrency anomalies (3 marks)
- Durability guarantees through WAL / redo logs (1 mark)
- Clear comparison with BASE / eventual consistency models in distributed architectures (2 marks)